

Atanu Saha

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PROFESSIONAL SUMMARY

Backend Software Engineer specializing in Python/FastAPI with ≈ 3 years of experience designing async, event-driven microservices and distributed systems. Built scalable SaaS platforms with APIs handling $\approx 40k +$ requests/day using NATS, PostgreSQL, and multi-cloud infrastructure (GCP & AWS). Skilled in API design, production operations, and automated testing.

PROFESSIONAL EXPERIENCE

Software Engineer

October 2023 – Present

Cefalo Bangladesh Ltd. — Backend Consultant for [TicketCo AS](#)

Dhaka, Bangladesh

- Architected and developed event-driven microservices (**FastAPI/FastStream/NATS**) across **3** services using **async/await** patterns; sustained **40k+ req/day** with $\approx 200ms$ average response time (**New Relic, 30-day rolling window**).
- Reduced NATS publish time spikes from **0.13–254s** to **30–40ms** by implementing **connection pooling** and tuning client configuration; eliminated multi-second latency spikes affecting booking flow reliability (measured via in-app timing logs, **10** runs).
- Refactored refund API to **asynchronous processing** with proper error handling and retries, cutting p95 response time by $\approx 72\%$ (from $\approx .5s$ to $\approx 700ms$); improved user experience and reduced server load.
- Owned production operability: incident triage, runbooks, and postmortems; improved observability with **structured logging, correlation IDs**, and health indicators; hardened failure handling with timeouts, retries, and circuit breakers.
- Achieved $\approx 85\%+$ test coverage using **PyTest** and **Cucumber BDD** with unit, and integration tests; enforced via GitHub Actions **CI** with coverage gates and automated validation.

Competitive Programming Trainer, Problem Setter & Judge

September 2022 – March 2024

American International University-Bangladesh (AIUB)

Dhaka, Bangladesh

- Led technical training in **C, C++, Data Structures, and Algorithms** for **100+** students, coordinating sessions, designing contests, and analyzing result based performance. Served as problem setter and judge for university contests, including [Intra AIUB 2024](#) and [AIUB CS Fest](#), organizing both online and offline events.

TECHNICAL SKILLS

Languages: Python, C++, SQL

Frameworks & Libraries: FastAPI, FastStream, SQLAlchemy, PyTest, Cucumber, asyncio

Data Stores & Messaging: PostgreSQL, NATS, SQLAlchemy ORM

Cloud & DevOps: GCP Cloud Run, AWS ECS, Docker, GitHub Actions

Observability: New Relic, Google Log Explorer, SolarWinds Papertrail

LLMs & genAI: RAG, LangChain, OpenAI, Hugging Face, Ollama, ChromaDB

PUBLICATIONS

A Comparative Analysis on Fake News Detection Methods

March 2022

International Conference on Computing Advancements (ICCA '22)

[ACM Digital Library](#)

EDUCATION

American International University-Bangladesh

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering — CGPA: 3.98/4.00

May 2018 – December 2021

ACHIEVEMENTS

- **Summa Cum Laude (Gold Medal)** – Highest Academic Honor, 22nd Convocation of AIUB, 2025
- **Dr. Anwarul Abedin Leadership Award** – Leadership Excellence, 22nd Convocation of AIUB, 2025
- **Honorable Mention** – 2022 ICPC Asia West Continent Final Contest (Team: AIUB.Singularity)
- **11th Place** – 2022 ICPC Asia Dhaka Regional Contest (Team: AIUB.Singularity)
- **18th Place** – 2021 ICPC Asia Dhaka Regional Online Preliminary Contest (Team: AIUB.Convergents)
- Competed in 20+ international and national programming contests – [ICPC Profile](#)